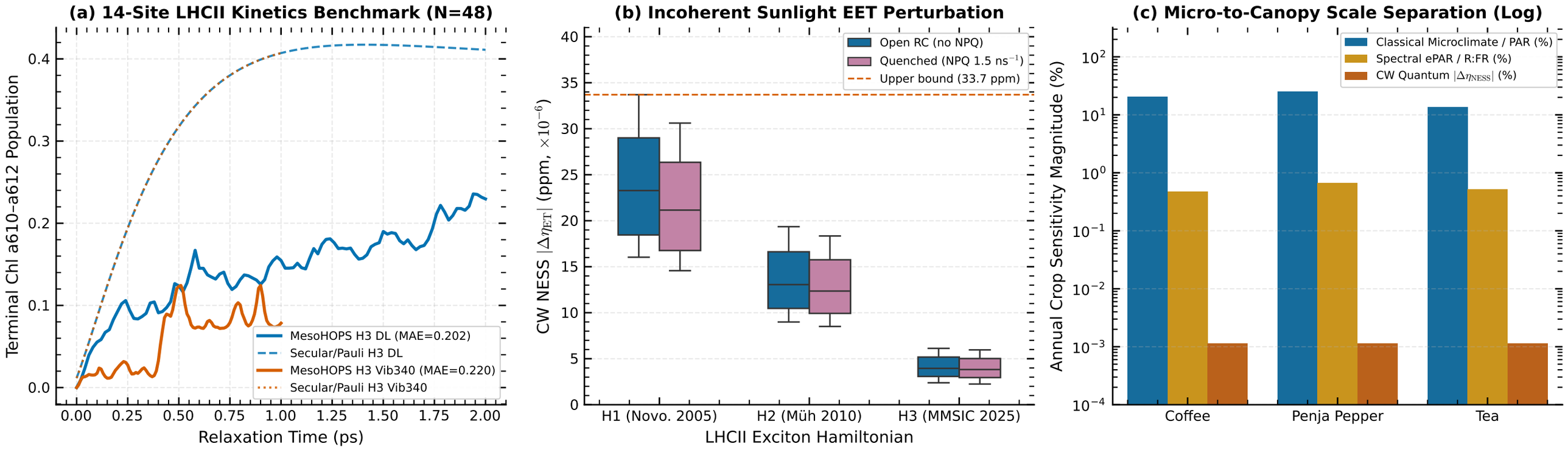
(a) 14-Site LHCII Kinetics Benchmark (N=48)
Terminal Chl a610–a612 Population
Relaxation Time (ps)
MesoHOPS H3 DL (MAE=0.202)
Secular/Pauli H3 DL
MesoHOPS H3 Vib340 (MAE=0.220)
Secular/Pauli H3 Vib340
(b) Incoherent Sunlight EET Perturbation
CW NESS $|\Delta\eta_{\mathrm{ET}}|$ (ppm, $\times 10^{-6}$)
Open RC (no NPQ)
Quenched (NPQ 1.5 ns^{-1})
Upper bound (33.7 ppm)
H1 (Novo. 2005)
H2 (Müh 2010)
H3 (MMSIC 2025)
LHCII Exciton Hamiltonian
(c) Micro-to-Canopy Scale Separation (Log)
Annual Crop Sensitivity Magnitude (%)
Classical Microclimate / PAR (%)
Spectral ePAR / R:FR (%)
CW Quantum $|\Delta\eta_{\mathrm{NESS}}|$ (%)
Coffee
Penja Pepper
Tea